

AIGR Deep Research Report —

Starmmerella bombicola Mdr / SbSLMdr.1

(UniProt H6TB12)

Focus: function_assignment **Hypothesis slug:** sophorolipid-export-and-mitochondrial-peptide-capacity **Source:** genes/STABO/mdr/mdr-ai-review.yaml (free-text)

Executive Judgment

Verdict: Sophorolipid-export role = **SUPPORTED (primary, experimentally established)**. Additional mitochondrial-inner-membrane localization + ATP-dependent oligopeptide-export claim = **OVER-ANNOTATED / weakly supported → recommend treating as non-core (candidate for removal or NOT qualifier)**.

The seed hypothesis asks whether H6TB12 genuinely has a second, mitochondrial oligopeptide-export function in addition to sophorolipid export. The evidence resolves cleanly:

1. **The sophorolipid-export function is the only experimentally supported activity/location.** UniProt annotates it with IMP evidence (GO:0140377 sophorolipid biosynthetic process; GO:0005886 plasma membrane), anchored to the deletion phenotype in

P 23516968

and reinforced by the transportome study

P 34998388

2. **The three mitochondrial/peptide GO terms are ALL IEA:TreeGrafter** — electronic, phylogenetically inferred annotations, not primary evidence. They are inherited at the family level from PANTHER **PTHR43394** ("ATP-DEPENDENT PERMEASE MDL1, MITOCHONDRIAL"), a family named after the yeast half-transporter Mdl1.
3. **Transporter architecture is diagnostic and refutes a genuine mitochondrial role.** H6TB12 is a **full-length** ABCB transporter (1299 aa, 12 TM helices, 2 NBDs; UniProt subfamily "Multidrug resistance exporter, TC 3.A.1.201") — the same structural class as

human P-glycoprotein/ABCB1 (1280 aa, 12 TM, 2 NBD). Genuine mitochondrial peptide exporters (yeast Mdl1 P33310 = 695 aa/1 NBD; human ABCB10 Q9NRK6 = 738 aa/1 NBD) are **half-size, single-NBD** proteins of a distinct "Mitochondrial peptide exporter" subfamily. Full-length MDR exporters are plasma-membrane multidrug/lipid pumps; mitochondrial inner-membrane peptide export is exclusive to the half-size Mdl1/ABCB10 subfamily.

Most important caveats. Absence of a positive mitochondrial assay does not formally prove absence of a minor moonlighting role ("inferred plasma-membrane localization alone does not prove every additional function absent," per the seed). However, curation standards weigh *positive experimental localization/function* (plasma membrane, sphorolipid export) against *family-level electronic inference contradicted by architecture*. On that basis the mitochondrial/peptide terms should not be presented as established functions of this protein. Generic ABC/ATPase/transmembrane-transport MF terms remain fully compatible and should be retained.

Evidence Matrix

#	Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context
1	P 23516968 (Van Bogaert 2013)	Mutant phenotype (deletion)	Supports sophorolipid-export role; competing with mito claim	Is Mdr a sophorolipid exporter?	Transporter knockout secretes SLs at only ~10% of WT → required for efficient SL export; part of SL biosynthetic gene cluster	<i>S. bombicola</i> , in vivo
2	P 34998388 (Claus 2022, transportome)	Mutant phenotype + comparative genomics	Supports; qualifies (redundancy)	Which transporters drive SL export?	ABC superfamily is "the main driver behind the highly efficient sophorolipid export"; identifies a second SL transporter outside the cluster (redundancy → SbSLMdr.2); distinguishes full-length MDR from half-size transporters	<i>S. bombicola</i> , in vivo
3	UniProt H6TB12 record	Database (feature/annotation provenance)	Qualifies/Refutes mito claim	Evidence class of each GO term	Mito/peptide terms (GO:0005743, GO:0015421, GO:0090374) are all IEA:TreeGraft ; SL export (GO:0140377) and plasma membrane (GO:0005886) are IMP:UniProtKB	Curated record
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#	Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context
	UniProt feature comparison (computed)	Structural/ evolutionary (architecture)	Refutes mito peptide-exporter identity	Is H6TB12 a half-size mito transporter?	H6TB12 = 1299 aa, 12 TM, 2 NBD (full-length), MDR-exporter subfamily = same class as ABCB1/P-gp (1280 aa, 12 TM, 2 NBD)	Sequence/ domain
5	P 11251115 (Young 2001, <i>Science</i>)	Direct assay + localization (of Mdl1)	Competing/ contextual	What protein does mito peptide export?	Mdl1 (a half -transporter in the mitochondrial inner membrane) exports ~600–2100 Da peptides from the matrix	<i>S. cerevisiae</i>
6	P 16710701 (Herget & Tampé 2007, review)	Review/ synthesis	Competing/ contextual	Which ABCB members are mitochondrial peptide exporters?	ABCB8/ABCB10 target inner mito membrane; MDL1 (yeast ABCB10 homolog) exports peptides — all half -transporters	Human/ yeast
7	P 15247210 (Galluhn & Langer 2004)	Direct assay (Mdl1 assembly)	Contextual	Mdl1 mechanism	Mdl1 half-transporter dimerizes and docks on F1F0-ATP synthase in inner membrane	<i>S. cerevisiae</i>
8	P 32474798 (Jezierska 2020)	Mutant phenotype	Contextual	<i>S. bombicola</i> mito carriers	<i>S. bombicola</i> 's characterized mitochondrial transporters are SLC-type citrate	<i>S. bombicola</i>

#	Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context
					carriers (SbCtp1/ SbYhm2), not the SL-MDR	
9	QuickGO / GO_REF:0000118 (computed, Iter 3)	Computational (annotation provenance)	Refutes/ qualifies mito claim	Independent support for mito terms?	GO:0015421, GO:0090374, GO:0005743 all share one provenance: IEA/ ECO:0007826/ GO_REF:0000118/ TreeGrafter, with- from = node PTN008681462; no other evidence line	Curated annotation graph

GO Curation Implications (leads — require curator verification)

GO term	Aspect	Current evidence	Lead recommendation
GO:0140377 sophorolipid biosynthetic process	BP	IMP:UniProtKB (P 23516968)	Retain (core). Consider also a transport BP (e.g., GO:0140161 monocarboxylic-acid/glycolipid export or export across plasma membrane) to capture the direct transport activity.
GO:0005886 plasma membrane	CC	IMP:UniProtKB	Retain (core); note localization is inferred from secretion phenotype, not microscopy.
GO:0005524 ATP binding / GO:0016887 ATP hydrolysis / ABC transmembrane transporter activity	MF	IEA:InterPro	Retain (compatible, well-supported by domains).
GO:0015421 ABC-type oligopeptide transporter activity	MF	IEA:TreeGrafter only	Generalize or remove for this protein. No peptide-transport assay; substrate is sophorolipid. Prefer a generic ABC-type transmembrane transporter activity term over the oligopeptide-specific one.
GO:0090374 oligopeptide export from mitochondrion	BP	IEA:TreeGrafter only	Remove / NOT qualifier (non-core). Architecture (full-length MDR) contradicts; no mitochondrial assay.
GO:0005743 mitochondrial inner membrane	CC	IEA:TreeGrafter only	Remove / NOT qualifier (non-core). Conflicts with experimental plasma-membrane localization; full-length MDR class is not mitochondrial.

Avoiding "protein binding" as a recommendation: the informative supported MF is **ABC-type transmembrane transporter activity coupled to ATP hydrolysis** (exporting sophorolipids), not oligopeptide transport.

Mechanistic Scope

- **Immediate molecular function (direct):** ATP-driven ABC-type efflux transport. H6TB12 uses two nucleotide-binding domains to hydrolyze ATP and translocate **acidic acylated/non-acylated sophorolipids** across the **plasma membrane** into the extracellular space (P 23516968).
- **Downstream / process level:** contributes to sophorolipid secretion; loss reduces (does not abolish) export because of paralog redundancy with SbSLMdr.2 (P 34998388) and alternative routes (P 23516968).
- **Claimed second function (mitochondrial oligopeptide export):** would require inner-mitochondrial-membrane targeting, matrix-facing peptide substrate handling, and the half-transporter/F1F0 assembly interface documented for Mdl1 (P 15247210).
None is demonstrated for H6TB12; the claim rests entirely on family-level electronic inference. This is best characterized as a **mis-transferred family annotation**, not a downstream phenotype of the real function.

Computed Provenance (Iteration 2 — sequence/architecture check)

Analysis run: 5-mer sequence similarity + length-ratio of H6TB12 against full-length MDR exporters and half-size mitochondrial peptide exporters (UniProt REST sequences); PANTHER family metadata via InterPro API.

Protein	Length (aa)	Len ratio vs H6TB12	5-mer Jaccard to H6TB12	Class
H6TB12 (target)	1299	1.00	1.000	full-length target
P21439 human ABCB4	1286	0.99	0.024	full-length MDR
P08183 human ABCB1/P-gp	1280	0.99	0.020	full-length MDR
Q9NRK6 human ABCB10	738	0.57	0.015	half-size mito peptide exporter
P33310 yeast Mdl1	695	0.54	0.013	half-size mito peptide exporter
P33311 yeast Mdl2	773	0.60	0.013	half-size mito
O75027 human ABCB7	752	0.58	0.006	half-size mito

- **Length ratio is decisive:** H6TB12 (1299 aa) equals full-length MDR exporters (ratio ~1.0) and is ~1.8–2× the size of every half-size mitochondrial exporter (ratio ~0.54–0.60), consistent with a fused (TMD-NBD)×2 architecture.
- **Similarity direction:** mean 5-mer Jaccard to full-length MDR refs (0.022) > to half-size mito refs (0.012). *Caveat:* absolute k-mer Jaccard is low because the references are cross-kingdom (human/*S. cerevisiae*); k-mer Jaccard is a crude proxy, so this is corroborating, not primary — the decisive evidence remains the UniProt domain architecture (12 TM, 2 NBD) and the "Multidrug resistance exporter (TC 3.A.1.201)" subfamily assignment.
- **PANTHER PTHR43394** metadata confirmed: family name = "ATP-DEPENDENT PERMEASE MDL1, MITOCHONDRIAL" — i.e., the family is named for the mitochondrial half-transporter Mdl1, mechanistically explaining why the mito/peptide GO-slim terms were transferred to a full-length MDR exporter.

(Executed code and full numeric output are retained in the Iteration 2 execution log as provenance.)

Iteration 3 — GO annotation provenance (QuickGO), resolving the PTN008681462 graft question

Analysis run: QuickGO annotation API for H6TB12 (evidence code, GO_REF, assigned-by, with/from).

GO ID	Aspect	Term	Evidence	GO_REF	Assigned by	With/From
GO:0140377	BP	sophorolipid biosynthetic process	IMP (ECO:0000315)	P 23516968	UniProt	—
GO:0005886	CC	plasma membrane	IMP (ECO:0000315)	P 23516968	UniProt	—
GO:0005886	CC	plasma membrane	IEA	GO_REF:0000044	UniProt	keyword SL
GO:0005524	MF	ATP binding	IEA	GO_REF:0000002	InterPro	IPR003439...
GO:0016887	MF	ATP hydrolysis activity	IEA	GO_REF:0000002	InterPro	IPR003439/0
GO:0140359	MF	ABC-type transmembrane transporter activity	IEA	GO_REF:0000002	InterPro	IPR011527
GO:0055085	BP	transmembrane transport	IEA	GO_REF:0000002	InterPro	IPR011527
GO:0016020	CC	membrane	IEA	GO_REF:0000002	InterPro	IPR011527/0
GO:0015421	MF	ABC-type oligopeptide transporter activity	IEA (ECO:0007826)	GO_REF:0000118	TreeGrafter	PTN0086814
GO:0090374	BP	oligopeptide export from mitochondrion	IEA (ECO:0007826)	GO_REF:0000118	TreeGrafter	PTN0086814
GO:0005743	CC	mitochondrial inner membrane	IEA (ECO:0007826)	GO_REF:0000118	TreeGrafter	PTN0086814

Key result — the seed's version-discrepancy question is resolved. All three disputed mitochondrial/oligopeptide terms carry the identical provenance: [IEA / ECO:0007826 / GO_REF:0000118 / TreeGrafter](#), with-from = **PTN008681462** — exactly the graft node the seed flagged. They rest on **one** automated phylogenetic graft with **no** independent supporting

evidence line. If that graft is unstable across PANTHER releases (the seed reports the current tree resolves PTN008681462 as an *Embryophyta* subtree with H6TB12 absent as an exact leaf), then **all three terms lose their sole support simultaneously** and can be treated as a single removable/NOT-qualifiable unit. The experimental terms (IMP,

P 23516968

and the generic InterPro ABC/ATP/membrane terms (GO_REF:0000002) are entirely independent of that node and are unaffected.

(Executed code and full QuickGO output are retained in the Iteration 3 execution log as provenance.)

Conflicts and Alternatives

- **Paralog/family confusion (primary driver of the over-annotation).** PANTHER PTHR43394 is named for Mdl1 and pools full-length and half-size ABCB proteins. TreeGrafter transferred Mdl1's GO-slim (mito inner membrane, oligopeptide export) to H6TB12 despite the architectural class difference. The seed's own citation

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P 34998388

explicitly distinguishes full-length MDR proteins from mitochondrial half-size transporters.

- **Localization conflict.** Experimental evidence places the protein at the plasma/cell membrane (consistent with extracellular sophorolipid secretion), directly conflicting with the IEA mitochondrial-inner-membrane term.
- **Organism context.** In *S. bombicola*, the transporters experimentally tied to mitochondria are SLC-type citrate carriers (SbCtp1/SbYhm2;

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P 32474798

not the SL-MDR — no independent line links H6TB12 to mitochondria.

- **Version/graft discrepancy (seed's TreeGrafter note).** The seed reports that frozen node PTN008681462 in PTHR43394 SF11 now resolves as an *Embryophyta* subtree and that H6TB12 is not present as an exact leaf, while the current tree GO-slim still retains mito/peptide terms on that path and the current IBD/GAF places mitochondrial-peptide IBDs on other paths. I could not programmatically re-derive the internal TreeGrafter graft (PANTHER graft internals were not accessible in this run), so I report this conservatively:

the discrepancy means the mito/peptide transfer is an **unstable, version-dependent electronic inference**, further weakening its use as evidence. Broad membrane/ATP/ABC IBDs along the path remain compatible.

Knowledge Gaps

1. **No direct subcellular imaging of H6TB12.** Checked: UniProt localization is ECO:0000305 (inferred from secretion). Matters because it is the crux of the mito-vs-plasma-membrane conflict. Resolve with GFP/immunofluorescence or fractionation in *S. bombicola*.

2. **No peptide-transport assay.** Checked: neither

P 23516968

nor 34998388 tests peptide substrates. Matters because GO:0015421/0090374 assert peptide handling. Resolve with in vitro reconstituted transport or peptide-efflux assays.

3. **TreeGrafter graft provenance (PTN008681462) — largely RESOLVED (Iter 3).** Confirmed via QuickGO that all three disputed terms share the exact provenance IEA/ECO:0007826/GO_REF:0000118/TreeGrafter with-from = PTN008681462, so they stand or fall together on that single graft. What remains unchecked: the internal PANTHER PTHR43394 tree topology (frozen vs current) that would confirm the seed's report of an *Embryophyta* resolution and H6TB12's absence as an exact leaf — the raw tree/GAF was not machine-accessible in this run. Resolve by inspecting the current PTHR43394 phylogeny and IBD GAF along the recovered path.

4. **Functional status of SbSLMdr.2 and cross-complementation.** Matters for how much of the export phenotype is attributable to H6TB12 specifically.

Discriminating Tests

1. **Localization:** C-terminal fluorescent tag or subcellular fractionation of H6TB12 in *S. bombicola* → plasma membrane (predicted) vs mitochondrial inner membrane (mito hypothesis). Single most decisive experiment.

2. **Substrate specificity:** reconstituted-vesicle or whole-cell efflux assay comparing sophorolipids vs 600–2100 Da peptides (the Mdl1 substrate range). MDR hypothesis predicts sophorolipid/hydrophobic-drug transport, not peptide export.
3. **Architecture-aware phylogenetics:** align H6TB12 against full-length ABCB1-type vs half-size Mdl1/ABCB10 references; confirm it branches with full-length MDR exporters (predicted). This distinguishes the two subfamilies without wet-lab work.
4. **TreeGrafter re-grafting audit:** recompute the graft node for H6TB12 in the current PTHR43394 tree and check whether mito/peptide GO-slim terms are on the recovered ancestral path or only on divergent (e.g., Embryophyta/half-transporter) branches.

Curation Leads (require curator verification)

- **Action change:** Downgrade the three `IEA:TreeGrafter` mitochondrial/peptide terms (GO:0005743, GO:0015421, GO:0090374) to **non-core**; candidate for removal or a **NOT** annotation given experimental plasma-membrane localization and full-length MDR architecture. Retain sophorolipid-export (GO:0140377) and plasma membrane (GO:0005886) as core; retain generic ATP/ABC MF terms.

- **Candidate references (verify snippets):**

- **P** 23516968

 — "knocking out the transporter gene yields mutants still able to secrete sophorolipids, though only at levels of 10% as compared with the wild type, suggesting alternative routes for secretion" (core secretory role + redundancy).
- **P** 34998388

 — ABC superfamily "harbors the main driver behind the highly efficient sophorolipid export"; "identify a second sophorolipid transporter that is located outside the sophorolipid biosynthetic gene cluster" (redundancy SbSLMDr.2; separates full-length MDR from half-size mito transporters).
- **P** 11251115

P 16710701

 — establish that mitochondrial inner-membrane oligopeptide export is the function of the

half-size Mdl1/ABCB10 subfamily (source of the transferred terms), not of full-length MDR exporters.

- **Suggested curator question:** Is there any experimental (imaging, fractionation, or peptide-transport) evidence for a mitochondrial or peptide role of H6TB12, or does all such support trace solely to PANTHER PTHR43394 TreeGrafter transfer?
- **Suggested experiment:** Fluorescent-tag localization + sophorolipid-vs-peptide efflux assay (Discriminating Tests 1–2).

Bottom Line

The prior/AI-proposed mitochondrial-inner-membrane + ATP-dependent oligopeptide-export capacity for H6TB12 is a **family-level TreeGrafter over-annotation** inherited from the Mdl1-named PANTHER family, contradicted by the protein's full-length P-gp/MDR architecture and its experimentally supported plasma-membrane sophorolipid-export function. The deletion/redundancy studies establish the **secretory** role, not a mitochondrial one. Broad ABC/ATP/transmembrane-transport terms remain valid; the specific mitochondrial and oligopeptide terms should be treated as non-core (remove or NOT) pending direct localization/substrate evidence.